

curl & wget Cheat Sheet

Fetch web pages, download files, call APIs and test web servers from the command line.

Using it

`curl [options] URL` ← *no options = print the response body to the screen*

Option	Means	Option	Means
-o file	Save to this file name	-H 'Name: value'	Adds a request header
-O	Save using the name from the URL	-d 'a=1&b=2'	Sends data (makes it a POST)
-L	Follow redirects (301/302)	--json '{...}'	POSTs JSON (curl 7.82+)
-I	Headers only (a HEAD request)	-X PUT	Other method: PUT, DELETE...
-i	Show headers and then the body	-u user:pass	Log in (basic auth)
-v	Verbose: connection, TLS, headers	-w '%{http_code}'	Print status, timing, etc. after
-s / -S	Silent / but still show errors	--resolve host:443:IP	Use this IP for host, skip DNS
-f	Fail (exit 22) on HTTP 400+ errors	--connect-timeout 5	Give up connecting after 5 s
-k	Skips certificate checks. Testing only	-C -	Resume a broken download

Reading curl -v https://example.com

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* Host example.com:443 was resolved. ← * = curl telling you what it's doing (DNS worked)
* Connected to example.com (203.0.113.10) port 443 ← TCP works: not DNS, not a firewall
* TLSv1.3 (OUT), TLS handshake, Client hello (1): ← TLS handshake starts
* Server certificate:
*  subject: CN=example.com ← the name the cert is for
*  expire date: Nov 30 23:59:59 2026 GMT ← when it runs out
*  issuer: C=US; O=Let's Encrypt; CN=R11
*  SSL certificate verify ok. ← trusted and matches the name
> GET / HTTP/2 ← > = what curl sent
< HTTP/2 200 ← < = what the server replied; 200 = OK
< content-type: text/html
< content-length: 1256 ← then a blank line, then the page itself
```

HTTP status codes

Code	Means	Code	Means
200	OK (304 = not modified, use your cached copy)	404	Not found (429 = too many requests)
301 / 302	Moved: see the location header, use -L	500	Server error: the app failed, check its logs
400	Bad request: malformed data	502	Bad gateway: proxy can't reach the app
401	Not logged in: send credentials	503	Unavailable: overloaded or in maintenance
403	Forbidden: logged in, but not allowed	504	Gateway timeout: app too slow for the proxy

wget (simpler, built for downloading)

Command	Does	Command	Does
wget URL	Downloads, keeping the file name	wget -i list.txt	Downloads every URL in a file
wget -O name URL	Saves as name (-qO- = to screen)	wget -r -np URL	Recursive, never up to parent dirs
wget -c URL	Continues a broken download	wget --mirror -np URL	Copies a site; reruns fetch changes

Handy recipes

Command	Does
curl --json '{"name":"web1"}' https://api.example.com/hosts	POSTs JSON to an API
curl -sIL http://example.com grep -iE '^(HTTP location)'	Shows each redirect hop and its status
curl -vsI https://example.com 2>&1 grep -E 'expire issuer'	Shows when the certificate expires
curl --resolve example.com:443:203.0.113.10 https://example.com	Tests a server before DNS moves
curl -o /dev/null -s -w '%{http_code} %{time_total}s\n' URL	Status code and total time

Remember: in scripts use `curl -fsSL`: fail on HTTP errors, no progress bar but show errors, follow redirects.